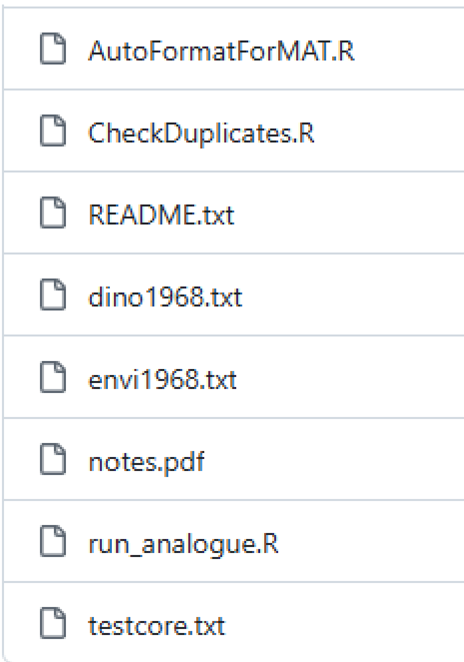
Part I: Getting started with the 'analogue' package

1. Install R and RStudio (optional). RStudio is an IDE (Integrated Development Environment) for R and Python. If you prefer to have a modern user interface with tools to help you be more productive, or if you have no idea at all, get RStudio. Download R and RStudio here: https://posit.co/download/rstudio-desktop/
2. Start RStudio. To install the 'analogue' package, type this in the console and press Enter: `install.packages('analogue')`
3. Various references are available online for the 'analogue' package. For example, the Rdocumentation page:
   https://www.rdocumentation.org/packages/analogue/versions/0.18.1,
   CRAN page with the official reference manual:
   https://cran.r-project.org/web/packages/analogue/index.html,
   and Gavin L. Simpson's published paper on the package:
   https://www.jstatsoft.org/article/view/v022i02.

Part II: Using the run_analogue.R script (example with dinocyst-based reconstruction using the n=1968 database)

1. **Download files and scripts from my Github repository**. Use this link https://downgit.github.io/#/home?url=https://github.com/xinerwu/MicroPal/tree/main/MAT to download a ZIP file of the subdirectory "MAT" that we will be using here. **Unzip** the ZIP file. You should have the following files in the "MAT" folder:

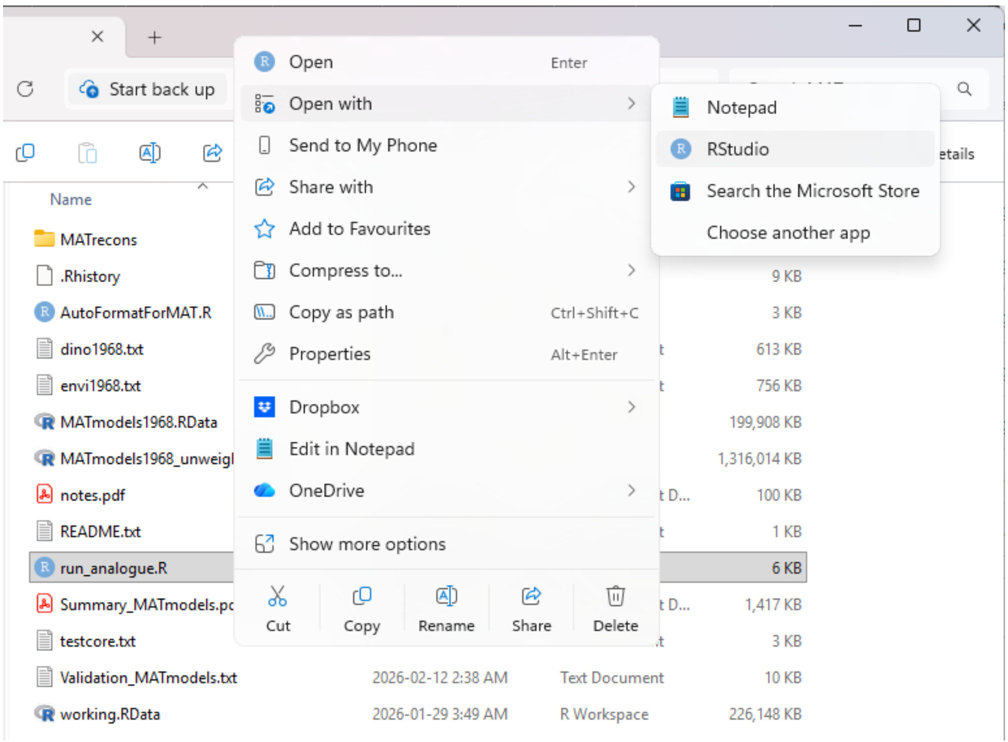


2. **Prepare the input file (your fossil proxy species data).**

If you are working with dinocyst: The column headers of your data file and the database file (dino1968.txt) have to be exactly the same and in the same order. Taxa abundances should be expressed in per mil (thus giving a sum of 1000). Your data file should be saved as tab-delimited text files. You can do this manually in Excel, or use the included AutoFormatForMAT.R script along with the instructions in notes.pdf.

If you are using another proxy (pollen, foraminifera, etc.): It does not matter if your data is in percentage [0 100], per mil [0 1000], or proportional [0 1]; however, your modern reference species data and your fossil species data must be on the exact same scale and have the exact same columns. To ensure that both data sets have the same set of columns, the analogue package includes a join function to merge the two data sets. But you can always do it manually if you are not familiar with R. My run_analogue.R script requires tab-delimited text files by default.

3. **Make sure your fossil data file is put into the "MAT" folder that you just unzipped.** Put it directly in "MAT". Do not create new subfolders. If you are using another proxy: you also need to put your modern reference species data and matching environmental variable data in two separate txt files in "MAT".
4. **Open the run_analogue.R script** (right-click – Open with... – Rstudio) with R/Rstudio.



5. **Modify the name of input file and output file, then save**.

If you are working with dinocyst, follow this:

```r
# Import modern reference species data file
spec <- read.delim('dino1968.txt',row.names=1)
# Import modern reference environmental variables file
envi <- read.delim('envi1968.txt',row.names=1)
# Import fossil species data file
core <- read.delim('testcore.txt',row.names=1)
# Name for output files:
output_file_name <- 'testcore'

# Change the number of analogues if wanted
n.analogues <- 5
# Choose the distance method
method <- "euclidean"
# Are you using dinocyst?
isDinocyst <- TRUE

# No bootstrapping by default, we use LOO for cross validation
# But if you do want to bootstrap, set bootstrap <- TRUE
# use n.boot=100 for testing/debugging, 1000 for formal analysis/publication
bootstrap <- FALSE
n.boot <- 100

# Reconstruct all variables (could take time, depends on the performance of your computer):
# target_env <- colnames(envi)
# Or specify environmental variables to be reconstructed:
target_env <- c('Ssummer','Swinter','Tsummer','Twinter','SeaIceC','SeaIcemonths','gCYC0217')

# Name of the file c
model_file <- 'MATmc
```

Replace testcore.txt by the name of your fossil data file

Decide the name for your output files

Add additional variables to reconstruct here, if wanted. Variable names should be the same as in envi1968.txt

If you are using another proxy, follow this:

```r
6  # Import modern reference species data file
7  spec <- read.delim('dino1968.txt',row.names=1)
8  # Import modern reference environmental variables
9  envi <- read.delim('envi1968.txt',row.names=1)
10 # Import fossil species data file
11 core <- read.delim('testcore.txt',row.names=1)
12 # Name for output files:
13 output_file_name <- 'testcore'
14
15 # Change the number of analogues if wanted
16 n.analogues <- 5
17 # Choose the distance method
18 method <- "euclidean"
19 # Are you using dinocyst?
20 isDinocyst <- TRUE
21
22 # No bootstrapping by default, we use LOO for cross validation
23 #                              bootstrap <- TRUE
24 #                               1000 for formal analysis/publication
25 bo
26 n.
27
28 # Reconstruct all variables (could take time, depends
29 # target_env <- colnames(envi)
30 # Or specify environmental variables to be reconstructed:
31 target_env <- c('Ssummer','Swinter','Tsummer','Twinter','SeaIceC','SeaIcemonths','gCYC0217')
32
33 # Name of the file containing MAT models that will be used/created
34 model_file <- 'MATmodels1968.RData'
```

Replace dino1968.txt by the name of your modern species data file

Replace envi1968.txt by the name of your modern environment data file

Replace testcore.txt by the name of your fossil data file

Decide the name for your output files

Change the distance method, e.g. “chord”

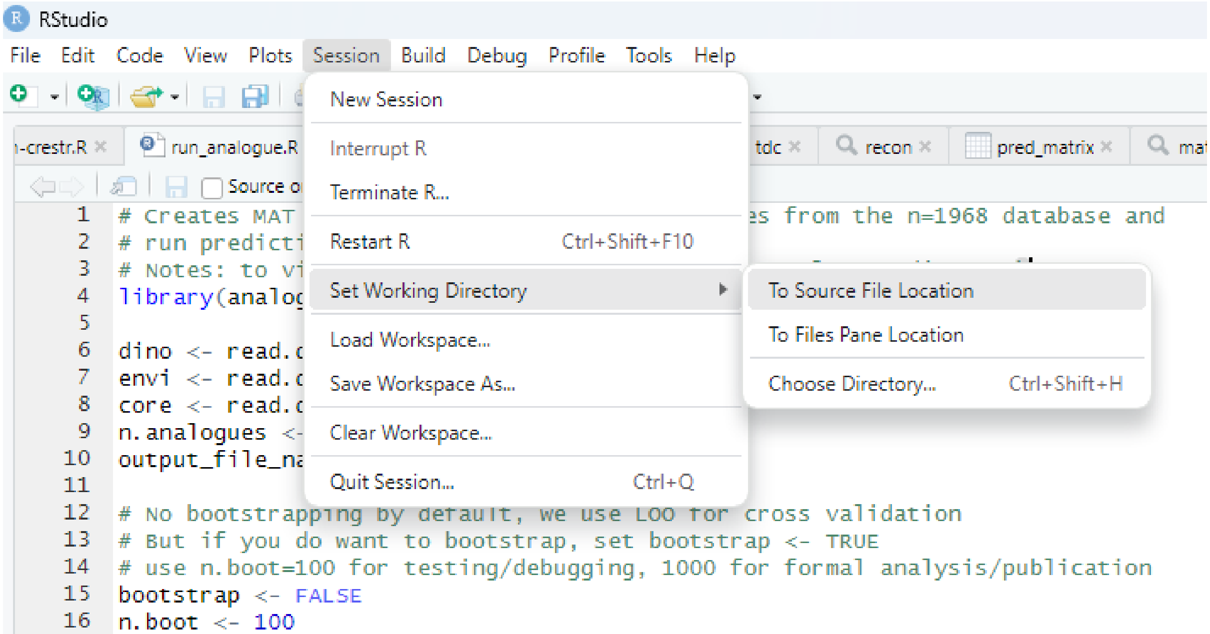
Change TRUE to FALSE since you are not using dinocyst

Simply comment out line 31 and uncomment line 29 to reconstruct all variables you have

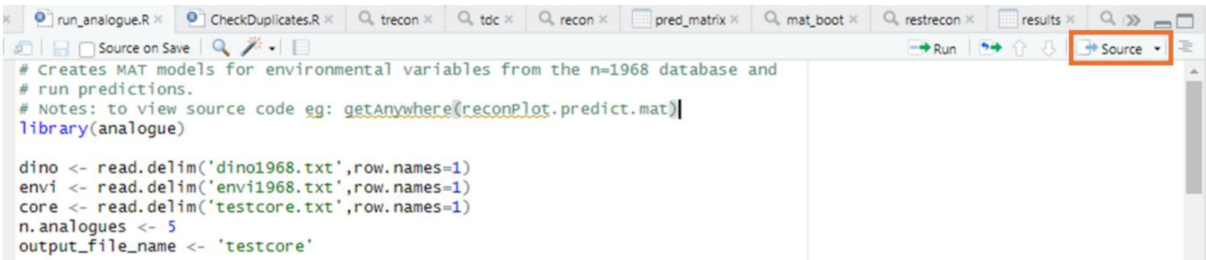
Or put your selection of environmental variables here. Variable names should be the same as in the envi file

Change the name of the MAT model file to something more descriptive of your modern calibration dataset

6. **Set working directory to the “MAT” folder**, so that R knows where to look for the input files and to save outputs. Go to Session – Set Working Directory – To source File Location.

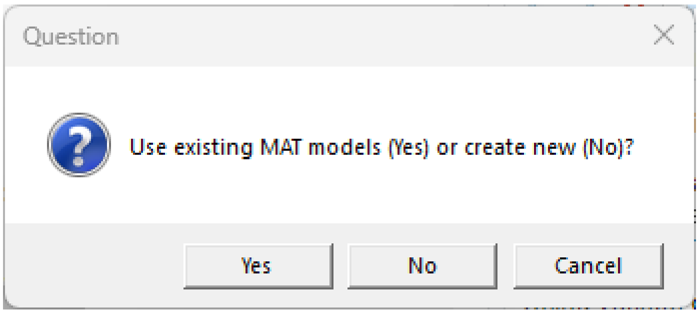


7. **If your system is in French**, with "," as the decimal symbol instead of "." and ";" as the list separator instead of "," and you don't want to change, here is some additional work for you. Find all the "write.csv" functions in the script and change them into "write.csv2". Similarly, if you wanted to use the AutoFormatForMAT.R script, you need to change "read.csv" to "read.csv2". If your tab-delimited text files also have "," as the decimal symbol, you also need to change "read.delim" to "read.delim2".
8. With your run_analogue.R script open in the window, **click on the Source button to run this script**. Some messages will appear in the Console window to inform you of the process.



9. **A pop-up window will appear**. The way the package 'analogue' works is that users need to first fit a MAT model to the training set, then use the fitted model to make predictions (reconstructions) from the fossil assemblage data. Since we normally don't make changes to our modern database, we can reuse the MAT models to run reconstructions on different cores to save time. When running the script for the very first time, new MAT models will be created no matter what you select. The MAT models will be saved in a file called MATmodels1968.RData (unless you changed the name) in the "MAT" folder. After that you can select Yes to reuse the models. If you

need to add additional variables to reconstruct, you need to create new models. Note that if you select No to create new ones, your previous MAT model file will be overwritten.



10. **Check the results**.

Model validation is automatically run when creating MAT models. The results are saved in the “MAT” folder. Summary_MATmodels.pdf contains graphics and Validation_MATmodels.txt contains the values of RMSEP and R^2 for each variable. With my script we use a weighted average of the analogues, so remember to check the statistics under “Inferences based on the weighted mean of k-closest analogues”. Note that these two files will also be overwritten when you create new models.

The reconstruction results are saved in an automatically created subfolder called “MATrecons”. XXX.csv contains the reconstructed values for the core. XXX_analogues.txt contains the closest analogues and their distances to each sample, as well as the percentiles of the dissimilarities for the training set (see Part III point 1 for explanation). XXX_preview.pdf contains stratigraphic plots of the reconstructions for a quick preview, as well as a plot of the minimum distance for each fossil sample (see Part III point 1 for explanation). Finally, XXX_ranges.csv contains the range (min & max) of the environmental values of the 5 closest analogues, which you could use as an indicator of uncertainty.

11. **Citations**.

If you use this software in your work, please cite the latest version archived on Zenodo:

Wu, X. (2026). R and MATLAB Scripts for Micropaleontological Proxy Data Processing (v1.0.0). Zenodo. https://doi.org/10.5281/zenodo.20494102

Citation for the analogue package is:

Simpson, G. L., & Oksanen, J. (2025). *analogue: Analogue matching and Modern Analogue Technique transfer function models, R package version 0.18.0. Retrieved from https://cran.r-project.org/package=analogue*.

Citation for the n=1968 dinocyst database (where the dino1968.txt, envi1968.txt, coor1968.txt files come from) is:

de Vernal A, Radi T, Zaragosi S, et al. (2020) Distribution of common modern dinoflagellate cyst taxa in surface sediments of the Northern Hemisphere in relation to environmental parameters: The new n=1968 database. *Marine Micropaleontology*. DOI: 10.1016/j.marmicro.2019.101796.

Citation for the pollen database (pol828.txt, env828.txt, cor828.txt) is:

Fréchette, B., de Vernal, A., Guiot, J., Wolfe, A. P., Miller, G. H., Fredskild, B., Kerwin, M. W., & Richard, P. J. H. (2008). Methodological basis for quantitative reconstruction of air temperature and sunshine from pollen assemblages in Arctic Canada and Greenland. *Quaternary Science Reviews*, *27*(11), 1197-1216. https://doi.org/https://doi.org/10.1016/j.quascirev.2008.02.016

Part III: Key differences between Joel Guiot's scripts and the 'analogue' package for running MAT reconstructions

1. The 'no analogue' situation

Joel Guiot's scripts apply a threshold when searching for analogues, meaning that when even the closest analogue to a sample is 'too far' according to this threshold, no reconstruction will be made on this sample. There is no such built-in threshold in the analogue package, the reconstruction will be run no matter what. Therefore, it is important to check the distance of the analogues using, for example, the built-in minDC function to extract the minimum distance for each fossil sample and plot it with the plot function (which is automatically done for you with my script). "A useful rule of thumb is that a fossil sample has no close modern analogues where the minDC for the sample is greater than the 5th percentile of the distribution of dissimilarity values for the training samples." (Simpson, 2007). See Figure 3 in Simpson (2007) for an example. The package also provides more sophisticated methods to determine the threshold for non-analogues, such as Monte Carlo resampling and ROC curves, which we will not discuss here.

Simpson GL (2007) Analogue Methods in Palaeoecology: Using the analogue Package. Journal of Statistical Software, Volume 22, Issue 2.

2. Using the logarithmic distance for dinocyst-based reconstruction

The analogue package does not have built-in logarithmic distance as a measure of dissimilarity, as opposed to Joel's scripts. Therefore, when using the analogue package for dinocyst-based reconstruction, we apply a logarithmic transformation on the species data then use the Euclidean distance to get the logarithmic distance. This is already coded in my script, but if you want to try a different type of distance with dinocyst data (chord, 'true' Euclidean, chi square, etc.), remember to remove the lines that do the logarithmic transformation. Note that my script only automatically applies a logarithmic transformation to dinocyst data, the other proxies are not affected.

3. Method of cross validation (for RMSEP)

Joel's scripts use, I believe, the holdout method (train-test split) where the data is randomly split into two parts: a training set and a testing set (e.g., 75% train, 25% test). The MAT model is trained once on the training set and evaluated on the testing set. The analogue package, by default, uses leave-one-out (LOO) cross validation to estimate the errors of prediction. if desired, it is also possible to perform bootstrap cross validation but computation time with 1000 bootstrap cycles can be very long (10+ minutes per environmental variable). Check lines 22-26 in run_analogue.R for bootstrap options.

4. Duplicates in your calibration dataset

Unlike Joel's scripts (which tolerate duplicates), the analogue package uses inverse distance weighting in the weighted MAT model. The weight given to an analogue is calculated as 1/distance. Therefore, during cross validation, if you have two samples with mathematically identical species composition (distance = 0), the weight becomes 1/0 and R cannot calculate the weighted average. I have added a safety check in my run_analogue.R script which removes repeated rows when detected, so that we can move forward instead of crashing. However, you should probably check the duplicate samples to decide whether they are accidental repetitions (simply remove them) or real replicates (consider averaging them into one sample, or jittering your species data if you must keep them). You can use my CheckDuplicates.R script to extract and inspect these samples.

5. Various

Joel's scripts require an older version of R (e.g., 3.4.0 works and is still available to download), other than that they are still good to use. I did a few tests which confirmed that Joel's scripts and the analogue package would generally yield identical results. One advantage of the analogue package is when you have a list of cores that you want to make reconstructions from, it is easy to write the reconstruction function into a loop and run it automatically through the list of cores (see https://github.com/xinerwu/MLD-Nlab/blob/ca4cbaf8d3bca48f6f248daa457dd3332075bf6c/MATreconstruction/run_predictions.R for an example).